1. The missing link budget

The selected empirical model is 9.46 dB short of 15 dB SNR for a 0 dBsm target at 1,000 m. The blue curve predicts +5.54 dB there. This report traces that result through reflector comparison, the Hallesaker walk, TX subset measurements and coherent RX reprocessing.

Budget term	Value
Walking baseline, normalized to 0 dBsm	C = 105.75 dB
Eight aligned TX: pair-based extrapolation	+13.49 dB
Coherent RX: selected rounded median	+6.30 dB
Combined empirical constant	C = 125.54 dB
Required constant: $15 + 40\log_{10}(1000)$	C = 135.00 dB
Additional total link gain required	9.46 dB

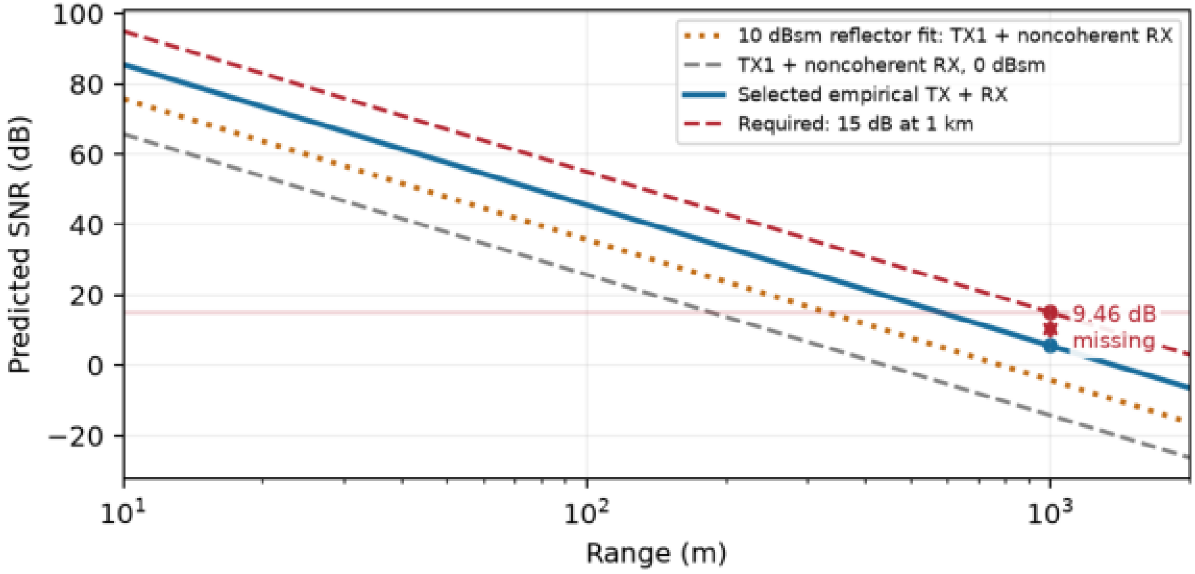

Figure 1. Orange: fitted 10 dBsm walking reflector, extrapolated beyond 15–51 m. Blue: selected 0 dBsm target estimate. The red-to-blue gap is 9.46 dB.

Model: $\mathrm{SNR}(R, \sigma) = 105.7465 + 13.4922 + 6.3 + \sigma(\mathrm{dBsm}) - 40\log_{10}(R/\mathrm{m})$. It predicts 15 dB at 580 m, 10 dB at 774 m and 0 dB at 1,376 m for a 0 dBsm target. The 15 dB criterion is a design target, not a measured probability-of-detection requirement.

The selected TX term is conditional: extrapolating +4.50 dB per doubling gives +13.49 dB for eight aligned TX. Using the actual eight-TX gain (+7.86 dB) instead predicts −0.10 dB at 1 km and leaves 15.10 dB missing. Thus the selected 9.46 dB gap is not demonstrated eight-TX performance.

An additional +10 dB total antenna-system gain would bring the selected model to 15.54 dB SNR at 1 km, only 0.54 dB above the target. Symmetric TX/RX antenna improvements would need about +4.73 dB per side to close 9.46 dB, subject to beam coverage. No propagation, implementation or fading margin is included.

2. Establishing the reflector reference

We compared the Hallesaker reflector with the nominal 10 dBsm reference in the lab. The recorded objects were the original reflector at 2.215 m and a second reflector to its left at 2.507 m. The radar rotated; the reflectors stayed fixed. The relative RCS result is measured, while adopting 10 dBsm for the walking reflector is a reference assumption, not an independent absolute calibration.

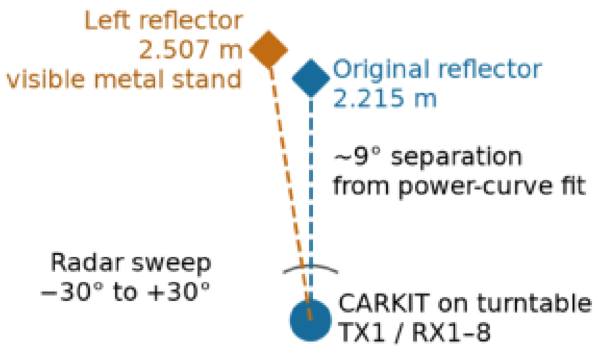

Figure 2. Top-view schematic, not surveyed geometry. The ~9° offset is inferred from the response curves. Reflector heights were not independently surveyed; the left object had a more visible metallic stand.

Reflector-comparison waveform	Saved configuration
Profile / enabled TX	rx-calibration-lab / TX1
Bandwidth / range-bin spacing	2.4996 GHz / 0.0600 m
Samples × ramps / PRI	512 × 1024 / 15.96 µs
Payload / prepayload / TX backoff	10.24 µs / 5.54 µs / 10 dB

The initial 400 MHz scan merged the returns: only one nearby stationary range peak appeared across 61 angles. The 2.5 GHz profile separated both targets. We scanned −30°:1°:+30°, waiting 0.5 s after motion and collecting 30 received frames per angle; each reflector was detected at every angle. Stationary-bin gates excluded adjacent Doppler copies.

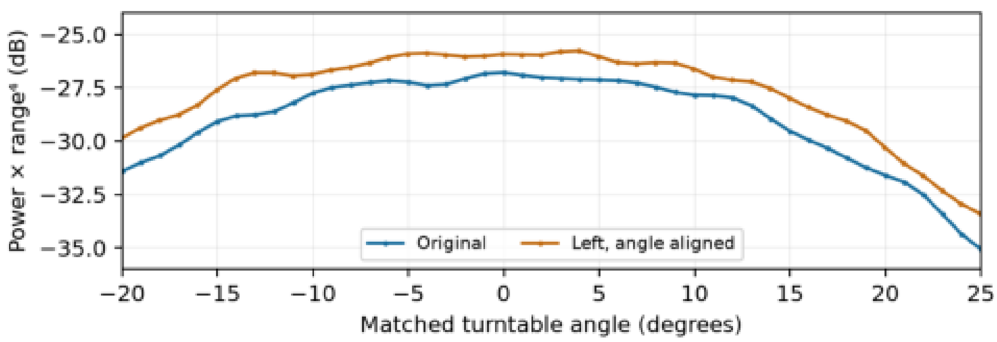

Figure 3. Range-corrected power after horizontal alignment; the remaining vertical offset estimates relative RCS.

We compared $P \times R^4$ and fitted a horizontal shift plus a constant level difference: −8.95° and +1.27 dB for the left object, with 0.32 dB RMS residual. Thus their apparent RCS values were close (ratio 1.34). The visible stand may contribute, but was not measured separately. We therefore adopted the user-specified 10 dBsm for the Hallesaker reflector, replacing 14 dBsm; we did not subtract a measured stand correction.

3. Walking the reflector at Hallesaker

The capture walk-hallesaker-tx1-1 contains 200 raw ADC frames. A corner reflector was carried away from and then back toward the radar. The saved track has 63 outbound and 74 inbound observations, spanning roughly 4.5–55 m overall. Time on the plot uses cadence inferred from range slope and Doppler, rather than assuming every recorded frame arrived at the configured 120 ms cycle.

Walking waveform	Saved configuration
Enabled TX / receivers	TX1 / RX1–8
Bandwidth / range-bin spacing	100.781 MHz / 1.48837 m
Samples × ramps / PRI	512 × 1024 / 15.96 µs
Payload / prepayload / carrier step	10.24 µs / 5.54 µs / 0 Hz
TX backoff / RX gain code	0 dB / 0

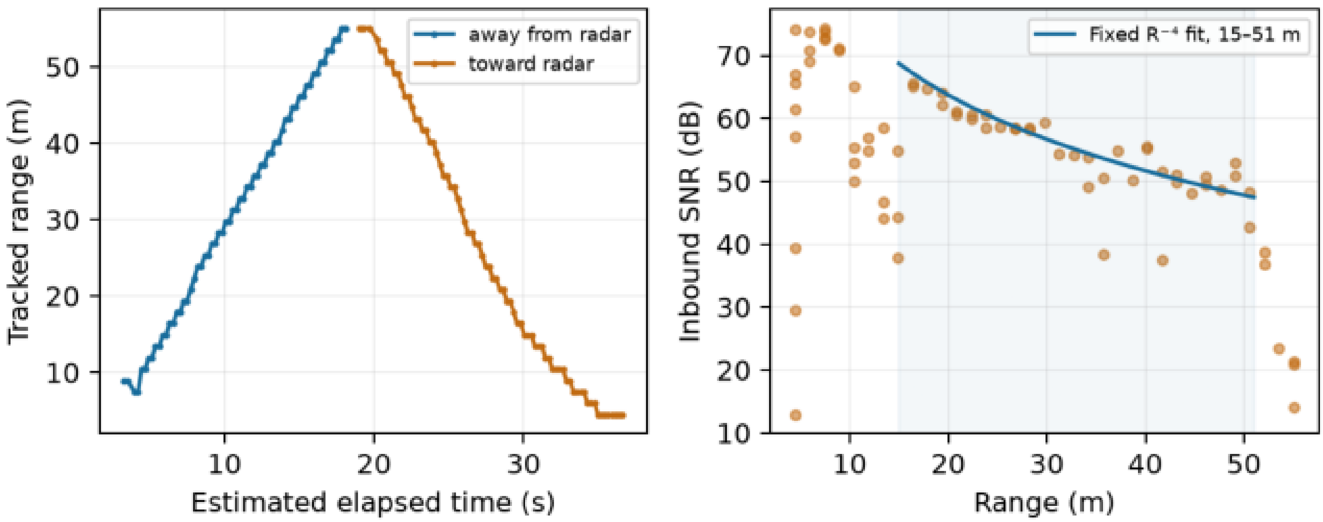

Figure 4. Extracted walking trajectory and inbound SNR. The fit uses maxima per populated range bin within 15–51 m.

Processing used the saved config.bin for dimensions and axes: Blackman range and Doppler windows, per-RX noise normalization, and an average of eight RX powers. No calibrated complex RX summation was present in this baseline. The single-TX effective phase length is 1; the full Doppler transform is retained.

The fixed-slope fit to 24 inbound maximum-per-range-bin samples gives $\mathrm{SNR} = 115.746526 - 40 \log_{10}(R/\mathrm{m})$, with 1.885 dB scatter. Subtracting the adopted 10 dBsm RCS gives C = 105.746526 dB. Changing 14 to 10 dBsm raised this normalized constant by 4 dB without changing the measured reflector curve.

This is an upper-envelope empirical range fit, not an average detection curve. The fit interval is only 15–51 m; projecting it to 1 km assumes R^{-4} propagation and equivalent target/antenna conditions. The page-1 range-budget model is for a 0 dBsm target; the observations and fitted curve above describe the adopted 10 dBsm walking reflector.

4. Updated TX measurements and extrapolation

Earlier DDMA calibration (calibration.toml, five frames, 2.60° RMS residual) gives TX1/6/7/8 correction phases 0°, −3.18°, +18.17°, +12.86°: a 21.35° cluster. We therefore tested TX1, TX1+6, TX1+6+7+8 and all eight. TX2/3 are far from this cluster (−138.79°, +123.69°), making partial cancellation plausible. These offsets were measured earlier, not reverified at 400 MHz or applied as fixed TX corrections.

TX-sweep waveform	Saved configuration
Bandwidth / range-bin spacing	399.61 MHz / 0.37537 m
Samples × ramps / PRI	512 × 1024 / 15.96 µs
Payload / prepayload / per-TX backoff	10.24 µs / 5.54 µs / 20 dB
TX codes / RX gain code	Identical zero increments / 2

All sweeps: −30°:1°:+30°, 0.5 s settling, 20 received frames per angle; strongest in-gate firmware SNR averaged in linear units over matching returns. Every angle had a return. Operation cycles were 120 ms for 1/2 TX and 50 ms for 4/8 TX; temporal sampling differs.

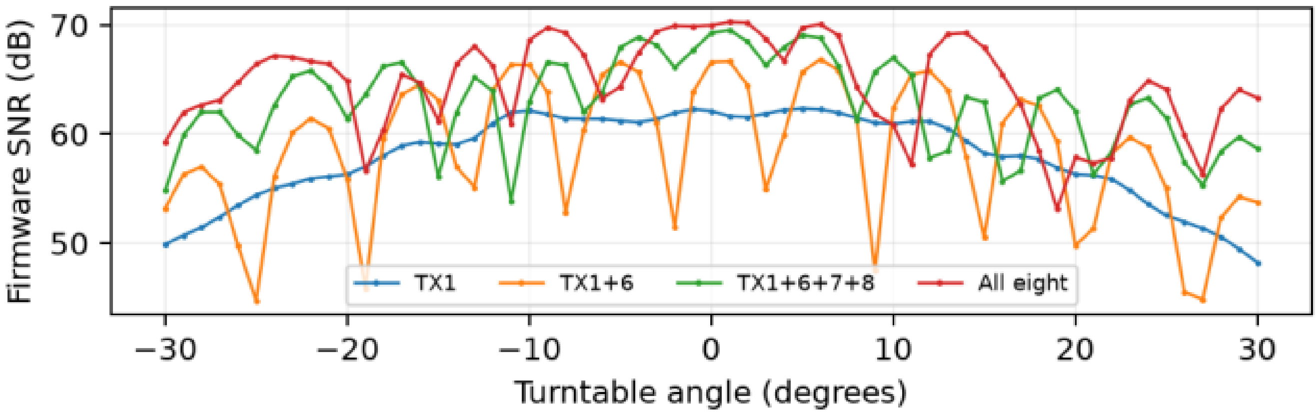

Figure 5. New 20 dB-backoff sweeps; the four-TX subset is now TX1/6/7/8.

Higher backoff reduced additional noise, without revealing missing signal gain. The raw results supersede interpreting the old +1.35 dB firmware result as signal-power gain. For the budget we use the new firmware pair gain: $G(N) = 4.4974 \log_2(N)$, hence $G(8) = 13.4922$ dB. Actual eight TX is 5.64 dB below that extrapolation. Phase mismatch is plausible, but the full gap is not proven recoverable; the raw/firmware SNR discrepancy remains unresolved.

5. TX-count gain and raw-data checks

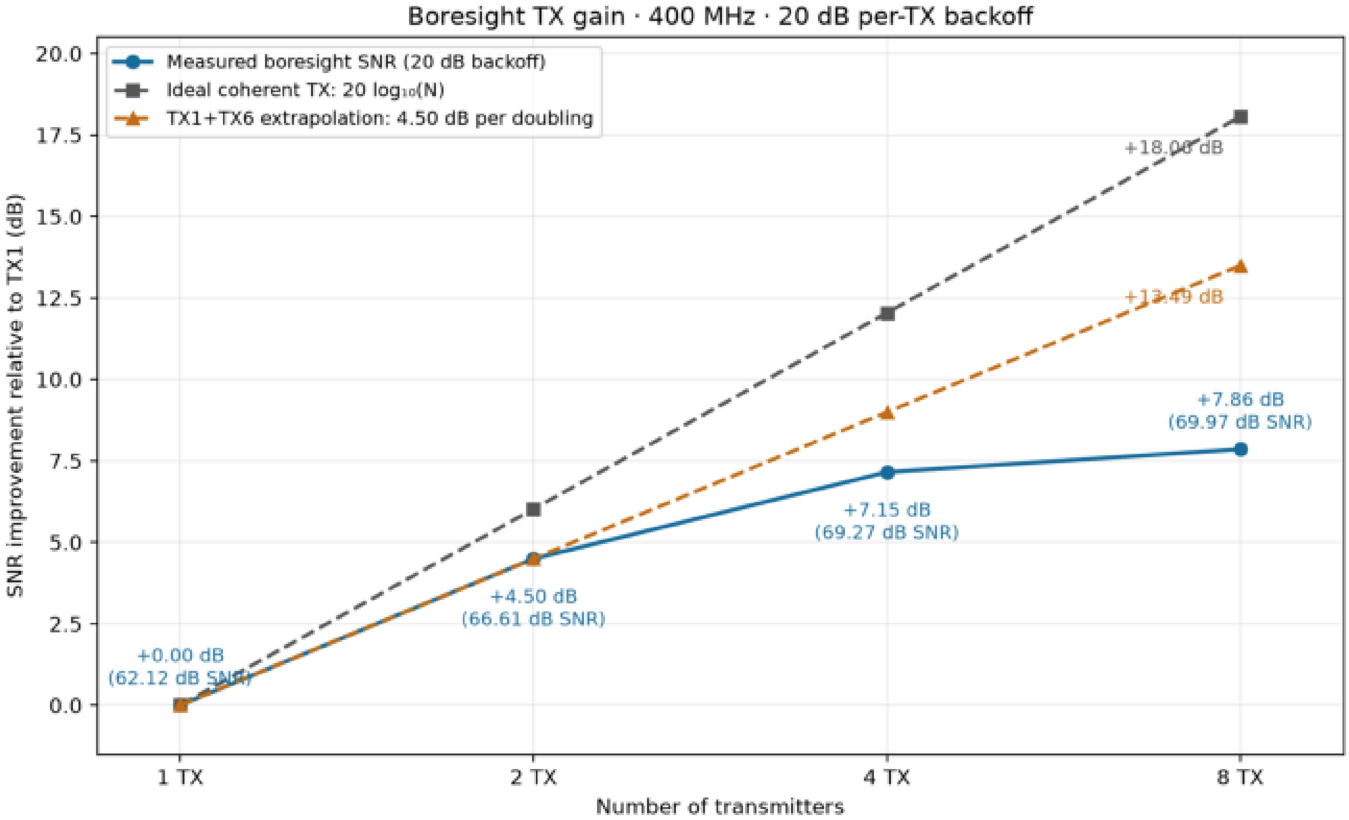

Figure 6. TX1 is normalized to 0 dB. Labels show gain, with measured firmware SNR in parentheses. The orange curve extrapolates the measured +4.50 dB per doubling; it is not a measured eight-TX result.

Raw boresight checks used ten analyzed ADC frames per condition, L2 Blackman FFTs and mean eight-RX power, without coherent RX summation or signal normalization. Noise was the farthest 20% of range, excluding |normalized Doppler| ≤ 0.05 (mean, not median). TX6 alone was within 0.06 dB of TX1 at 10 dB backoff. Pair signal gains were +5.79/+5.64 dB at 10/20 dB backoff; noise increases +1.16/+0.14 dB, giving raw SNR gains +4.63/+5.50 dB. No clipping; pair ADC peaks 47.4%/18.6%.

Raw signal power and firmware SNR are different measurements. At 20 dB backoff, raw pair signal gain was +5.64 dB and raw SNR gain +5.50 dB, versus +4.50 dB in the firmware angle sweep. These sequential measurements do not resolve that discrepancy. The range budget uses the explicitly selected firmware pair extrapolation, +13.49 dB for eight TX, and retains the measured-eight-TX alternative.

6. RX gain and the resulting requirement

We reprocessed all 137 saved walking-track frames twice at identical range–Doppler cells: the original eight-RX noncoherent average, and a complex RX sum using the fixed calibration-400.toml weights. The latter was normalized by the measured noise of the weighted sum, including cross-channel covariance. The recomputed baseline agrees with the original SNR values within 0.000007 dB.

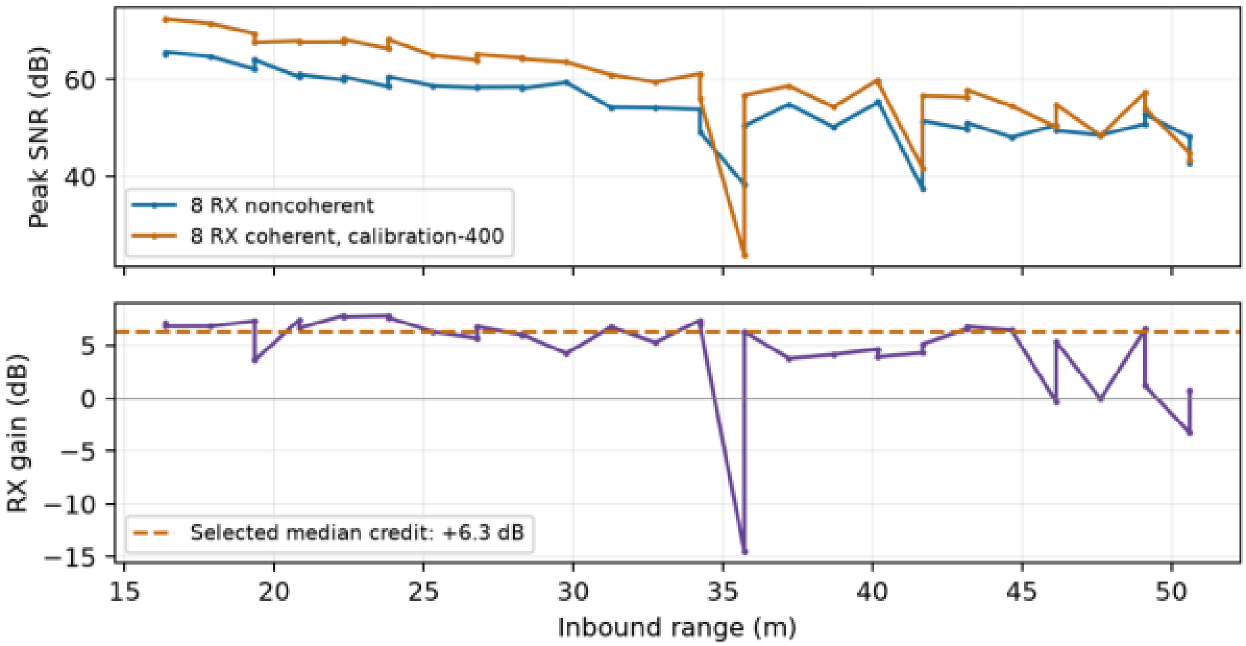

Figure 7. Paired inbound samples in the 15–51 m interval. Fades remain in the results; the selected median credit is +6.3 dB.

RX comparison	Measured extra SNR
Median, inbound 15–51 m / both legs	+6.28 dB / +6.29 dB
Mean on original 24 range-fit samples	+4.82 dB
User-selected typical-gain model credit	+6.30 dB
Ideal independent-noise RX benchmark	+9.03 dB

The range model deliberately uses the rounded median, +6.3 dB, rather than the lower +4.82 dB fit-sample mean. This is a typical tracked-return estimate, not protection against mismatches: 35 of 39 inbound fit-interval samples improve, and one loses 14.48 dB. A 3×3 target-patch check gives a similar median (+6.20 dB). No digital clipping was found in the processed frames.

The resulting budget is C = 105.75 + 13.49 + 6.30 = 125.54 dB, versus 135.00 dB required: 9.46 dB additional link gain. The actual eight-TX alternative leaves 15.10 dB missing. The RX term is measured on the walk; the selected TX term extrapolates a lab pair and assumes transfer from 400 MHz / 20 dB backoff to the ~101 MHz / 0 dB-backoff walk. Their additivity has not been demonstrated in a combined 1 km experiment. Reference RCS, elevation, multipath, beam coverage, phase stability and long-range propagation remain uncertainties.

Traceability: [1] two-reflectors-rcs-tx1-2500mhz-2026-09-11/rcs-estimate.json; [2] walk-hallesaker-tx1-1/reflector-snr-r4-fit.txt; [3] tx-count-20db-angle-sweeps-2026-09-11/tx-count-snr-improvement.json; [4] walk-hallesaker-tx1-1/rx-coherent-400-calibration/summary.json; [5] walk-hallesaker-tx1-1/empirical-snr-8tx-8rx-0dbsm.json. All paths are under data/. Raw TX checks: tx-boresight-raw-2026-09-11/backoff-comparison.json. Radar equation: mathworks.com/help/radar/ug/radar-equation.html.